Muddasir Rahim

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Professional Summary

- Highly motivated individual with experience in assisting with research projects. Proven track record of publishing in prestigious IEEE journals and conferences. Skilled in Matching Theory with proficiency in MATLAB and Python programming, adept at implementing complex algorithms and conducting quantitative analysis. Dedicated to advancing research in telecommunications and contributing impactful solutions to technological challenges.

Education

2020 – 2025

Ph.D. in Electrical Engineering, CGPA: 3.8/4.3, ÉCOLE DE TECHNOLOGIE SUPÉRIEURE UNIVERSITÉ DU QUÉBEC, Montreal, Canada.
Thesis title: *Intelligent Reconfigurable Surface-Assisted Terahertz Communications," under supervision of* ***Prof. Georges Kaddoum****.*
In my Ph.D., I am working on Intelligent Reconfigurable Surface (IRS) assisted Terahertz (THz) communication. In this regard, I am working on resource management in IRS-assisted THz networks in multi-user scenarios. Moreover, four journals and one conference article are accepted at the IEEE venue for publication.

2018 – 2020

M.Sc in Electrical Engineering, CGPA: 3.82/4, COMSATS University Islamabad, Islamabad, Pakistan.
Thesis title: *Efficient Radio Spectrum Utilization and Mobility in CRN for Improved Quality of Service, under supervision of* ***Prof. Riaz Hussain****.*
During my master's degree thesis, I worked on spectrum sensing and sharing in cognitive radio networks. I have published two research articles from my thesis. Furthermore, I have published three journal articles while working on different research projects in collaboration with different Professors at COMSATS University.

2012 – 2016

B.S in Electrical (Telecommunication) Engineering, COMSATS University Islamabad, Islamabad, Pakistan.
Thesis title: *Energy Efficient Wireless Sensor Networking for a Remote Control Patient, under supervision of* ***Prof. Mustafa Shakir****.*
During my bachelor's degree, I designed Medium Access Control (MAC) for Wireless Body Area Networks (WBANs) in which we used CSMA/CA with different Contention Window (CW) size for variable nodes with different priority. In this regard, the nodes with the highest priority were assigned minimum CW size, however, less priority nodes were assigned a higher value of CW. Consequently, published one journal article.

Employment History

2025 —-

Postdoctoral Fellow, École Polytechnique Montréal, under the supervision of Prof. Soumaya Cherkaoui.
Postdoctoral Fellow at École Polytechnique Montréal, focusing on efficient resource allocation for intelligent reconfigurable surface-assisted integrated terrestrial and non-terrestrial networks. My broader research interests include Cognitive Radio Networks (CRN), Device-to-Device (D2D) communication, Wireless Sensor Networks (WSN), Industrial IoT, coexistence of enhanced Mobile Broadband (eMBB) and Ultra-Reliable Low-Latency Communications (URLLC), Intelligent Reconfigurable Surfaces (IRSs), Matching Theory, and Terahertz (THz) communications.

2019 –2020

Research Assistant, COMSATS University Islamabad, HEC NRPU project, under the supervision of Prof. Riaz Hussain.
During my master's degree, I worked as a Research Assistant on HEC funded project. During this research, I worked on spectrum sensing and spectrum sharing in cognitive radio networks. Furthermore, I published two journal research articles during this project.

2016

Network Engineer, National Telecommunication Corporation Pakistan.
Worked at the National Telecommunication Corporation for six months as a network engineer.

Research Publications

Journal Articles

1. A. Basit, **M. Rahim**, T. N. Do, N. Adam, and G. Kaddoum, "DRL-based maximization of the sum cross-layer achievable rate for networks under jamming," *IEEE Transactions on Network and Service Management*, 2025.
2. **M. Rahim**, A. Basit, B. Selim, and G. Kaddoum, "Hierarchical irs-assisted user association for sum-rate maximization in hybrid thz leo-haps-terrestrial networks," *IEEE Communications Letters*, pp. 1–1, 2025. DOI: 10.1109/LCOMM.2025.3619420.
3. **M. Rahim**, G. Kaddoum, and T. N. Do, "Joint Devices and IRSs Association for Terahertz Communications in Industrial IoT Networks," *IEEE Transactions on Green Communications and Networking*, vol. 8, no. 1, pp. 375–390, 2024. DOI: 10.1109/TGCN.2023.3332571.
4. **M. Rahim**, T. L. Nguyen, T. N. Do, and G. Kaddoum, "Joint Power and User Allocation in Coexistence of eMBB and URLLC Services," *IEEE Communications Letters*, pp. 1–1, 2024. DOI: 10.1109/LCOMM.2024.3411399.
5. **M. Rahim**, T. L. Nguyen, and G. Kaddoum, "JPUSA in Coexistence of eMBB and URLLC Services in Multi-Cell IRS-Assisted Terahertz Networks," *IEEE Transactions on Green Communications and Networking*, pp. 1–1, 2024. DOI: 10.1109/TGCN.2024.3498834.
6. **M. Rahim**, T. L. Nguyen, G. Kaddoum, and T. N. Do, "Multi-IRS-Aided Terahertz Networks: Channel Modeling and User Association With Imperfect CSI," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 836–855, 2024. DOI: 10.1109/OJCOMS.2024.3353173.
7. A. Ranjha, G. Kaddoum, **M. Rahim**, and K. Dev, "URLLC in UAV-enabled multicasting systems: A dual time and energy minimization problem using UAV speed, altitude and beamwidth," *Computer Communications*, vol. 187, pp. 125–133, 2022. URL: https://doi.org/10.1016/j.comcom.2022.02.013.

8 A. Iqbal, **M. Rahim**, R. Hussain, *et al.*, "cDERSA: Cognitive D2D Enabled Relay Selection Algorithm to Mitigate Blind-Spots in 5G Cellular Networks," *IEEE Access*, vol. 9, pp. 89 972–89 988, 2021. DOI: 10.1109/ACCESS.2021.3090965.

9 M. Z. Khan, **M. Rahim**, M. A. Javed, F. Ghabban, O. Ameerbakhsh, and I. Alfadli, "A D2D assisted multi-hop data dissemination protocol for inter-UAV communication," *International Journal of Communication Systems*, vol. 34, no. 11, e4857, 2021. URL: https://doi.org/10.1002/dac.4857.

10 **M. Rahim**, S. Ali, A. N. Alvi, *et al.*, "An intelligent content caching protocol for connected vehicles," *Transactions on Emerging Telecommunications Technologies*, vol. 32, no. 4, e4231, 2021. URL: https://doi.org/10.1002/ett.4231.

11 **M. Rahim**, R. Hussain, I. L. Khan, *et al.*, "Self-Organized Efficient Spectrum Management through Parallel Sensing in Cognitive Radio Network," *Wireless Communications and Mobile Computing*, vol. 2021, no. 1, p. 5 552 012, 2021. URL: https://doi.org/10.1155/2021/5552012.

12 **M. Rahim**, A. S. Alfakeeh, R. Hussain, *et al.*, "Efficient channel allocation using matching theory for QoS provisioning in cognitive radio networks," *Sensors*, vol. 20, no. 7, p. 1872, 2020. URL: https://doi.org/10.3390/s20071872.

13 **M. Rahim**, M. A. Javed, and A. N. Alvi, "SDMA-based Distributed Device Discovery for D2D communication," *Turkish Journal of Electrical Engineering and Computer Sciences*, vol. 28, no. 6, pp. 3183–3192, 2020. DOI: 10.3906/elk-2002-113.

14 **M. Rahim**, M. A. Javed, A. N. Alvi, and M. Imran, "An efficient caching policy for content retrieval in autonomous connected vehicles," *Transportation Research Part A: Policy and Practice*, vol. 140, pp. 142–152, 2020. URL: https://doi.org/10.1016/j.tra.2020.08.005.

15 M. Shakir, O. U. Rehman, **M. Rahim**, *et al.*, "Performance optimization of priority assisted csma/ca mechanism of 802.15. 6 under saturation regime," *Sensors*, vol. 16, no. 9, p. 1421, 2016. URL: https://doi.org/10.3390/s16091421.

Conference Proceedings

1 **M. Rahim**, A. Basit, T. L. Nguyen, and G. Kaddoum, "Maximizing URLLC reliability through JPSA for URLLC services in IRS-aided Terahertz networks," in *2025 IEEE Wireless Communications and Networking Conference (WCNC)*, IEEE, 2025, pp. 1–6.

2 A. Basit, **M. Rahim**, G. Kaddoum, T. N. Do, and N. Adam, "DRL-based Dynamic Channel Access and SCLAR Maximization for Networks under Jamming," in *2024 IEEE Wireless Communications and Networking Conference (WCNC)*, 2024, pp. 1–6. DOI: 10.1109/WCNC57260.2024.10571262.

3 **M. Rahim**, T. L. Nguyen, G. Kaddoum, and T. N. Do, "User Association Optimization for IRS-Aided Terahertz Networks: A Matching Theory Approach," in *2024 IEEE Wireless Communications and Networking Conference (WCNC)*, 2024, pp. 1–6. DOI: 10.1109/WCNC57260.2024.10570866.

Skills

Languages	Strong reading, writing and speaking competencies for English.
Programing languages and tools	Matlab Python Omnet++ C, C++ CISCO Packet Tracer HFSS